



Graphs in Machine Learning

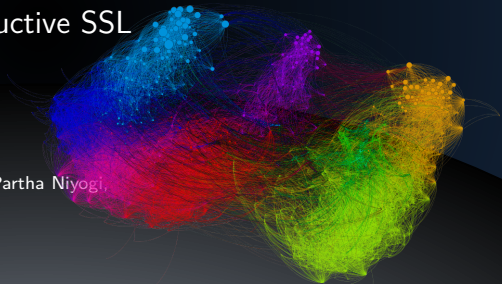
Inductive Generalization Bounds

Theoretical Guarantees for Inductive SSL

Michal Valko

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Partially based on material by: Mikhail Belkin, Partha Niyogi,
Olivier Chapelle, Bernhard Schölkopf



Inductive Generalization Bounds

We may want to bound the **risk**

$$R_P(f) = \mathbb{E}_{P(\mathbf{x})} [\mathcal{L}(f(\mathbf{x}), y(\mathbf{x}))]$$

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$$R_P(f) \leq \hat{R}_P(f) + \text{error terms}$$

Inductive Generalization Bounds

Using classical SLT tools (Equations 3.15 and 3.24 **vapnik1995nature**), with probability $1 - \eta$

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How to bound $\mathcal{L}(f(\mathbf{x}_i), y_i)$?

Inductive Generalization Bounds

For any $y_i \in \{-1, 1\}$ and ℓ_i^*

$$\frac{1}{N} \sum_i \mathcal{L}(f(\mathbf{x}_i), y_i) \leq \frac{1}{N} \sum_i \mathcal{L}(f(\mathbf{x}_i), \text{sgn}(\ell_i^*)) + \frac{1}{N} \sum_i (\ell_i^* - y_i)^2$$

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For any $y_i \in \{-1, 1\}$ and ℓ_i^*

$$\begin{aligned} \frac{1}{N} \sum_i \mathcal{L}(f(\mathbf{x}_i), y_i) &\leq \frac{1}{N} \sum_i \mathcal{L}(f(\mathbf{x}_i), \text{sgn}(\ell_i^*)) + \frac{1}{N} \sum_i (\ell_i^* - y_i)^2 \\ &\leq \left(\frac{1}{N} \sum_i \mathcal{L}(f(\mathbf{x}_i), \text{sgn}(\ell_i^*)) \right) + \hat{R}_P(\ell^*) + \Delta_T(\beta, n_I, \delta) \end{aligned}$$

Inductive Generalization Bounds

Combining **inductive** + **transductive** error

With probability $1 - (\eta + \delta)$.

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We need to account for ε . With probability $1 - (\eta + \delta)$.

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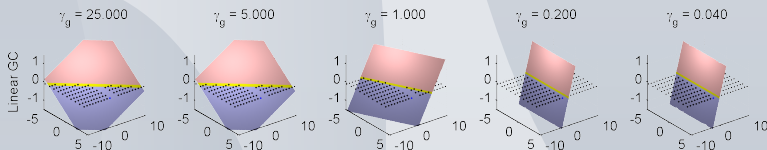
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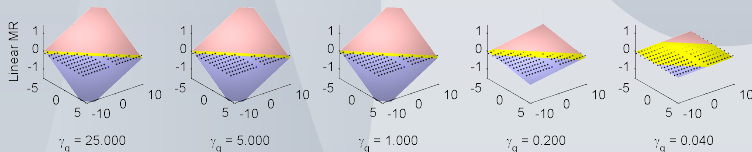
We should have $\varepsilon \leq n_I^{-1/2}$!

SSL with Graphs: LapSVMs and MM Graph Cuts

MMGC for 2D data and **linear** $\mathcal{K}$ works as we want

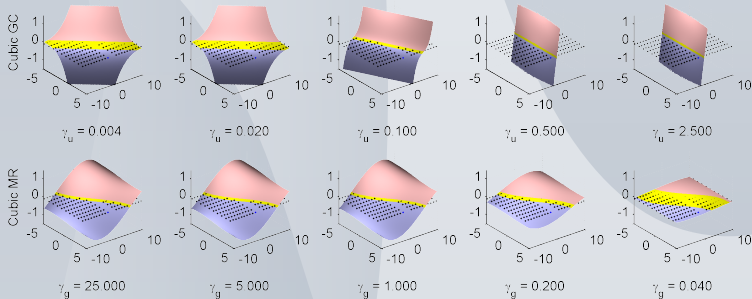


LSVM for 2D data and **linear** $\mathcal{K}$ only changes the slope



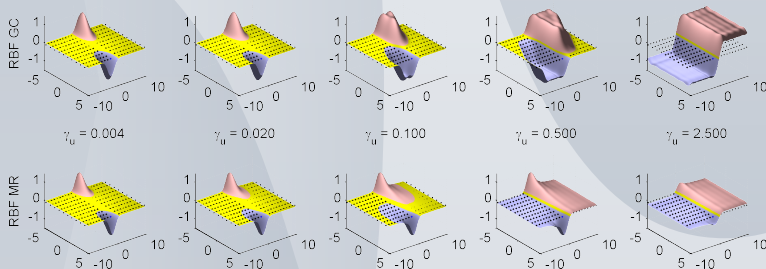
SSL with Graphs: LapSVMs and MM Graph Cuts

LSVM for 2D data and **cubic** $\mathcal{K}$ is also not so good



SSL with Graphs: LapSVMs and MM Graph Cuts

MMGC and LSVM for 2D data and RBF $\mathcal{K}$



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<https://misovalko.github.io/mva-ml-graphs.html>